

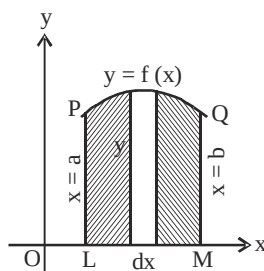
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APPLICATION OF THE INTEGRALS

KEY CONCEPT INVOLVED

Area Under Simple Curves

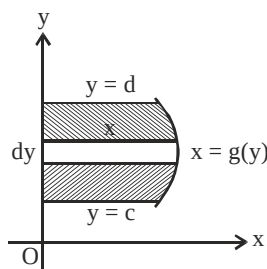
- Let us find the area bounded by the curve $y = f(x)$, x-axis and the ordinates $x = a$ and $x = b$. Consider the area under the curve as composed of large number of thin vertical strips let there be an arbitrary strip of height y and width dx . Area of elementary strip $dA = ydx$, where $y = f(x)$. Total Area A of the region between x-axis, ordinates $x = a$, $x = b$ and the curve $y = f(x)$ = Sum of areas of elementary thin strips across the region PQML



$$A = \int_a^b dA = \int_a^b ydx = \int_a^b f(x)dx$$

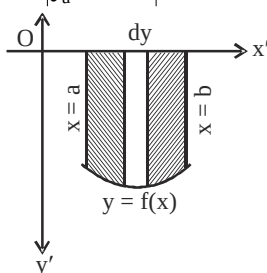
- The area A of the region bounded by the curve $x = g(y)$, y-axis and the lines $y = c$ and $y = d$ is given by

$$A = \int_c^d xdy$$



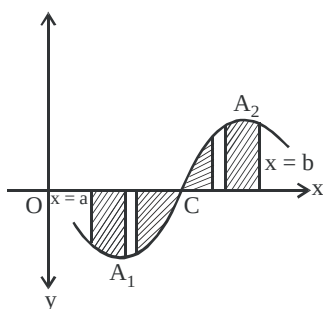
- If the curve under consideration lies below x-axis, then $f(x) < 0$ from $x = a$ to $x = b$, the area bounded by the curve $y = f(x)$, and the ordinates $x = a$, $x = b$ and x-axis is negative. But the numerical value of the area is to

be taken into consideration. Then Area = $\left| \int_a^b f(x)dx \right|$



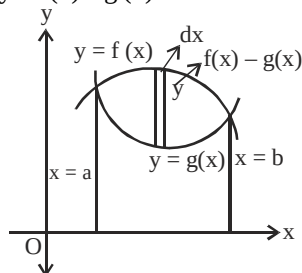
4. Let some portion of the curve is above x-axis and some portion is below x-axis. Let A_1 be the area below x-axis and A_2 be the area above of x-axis. Therefore Area bounded by the curve $y = f(x)$, x-axis and the ordinates $x = a$ and $x = b$.

$$A = |A_1| + A_2$$



Area between Two curves

5. Let the two curves be $y = f(x)$ and $y = g(x)$. Suppose these curves intersect at $x = a$ and $x = b$. Consider the elementary strip of height y where $y = f(x) - g(x)$ with width dx

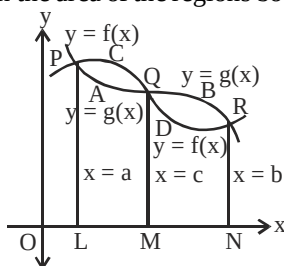


$$\therefore da = ydx$$

$$\Rightarrow A = \int_a^b (f(x) - g(x)) dx = \int_a^b f(x) dx - \int_a^b g(x) dx$$

i.e. $A = \text{Area bounded by the curve } y = f(x) - \text{Area bounded by the curve } y = g(x)$

6. If the two curves $y = f(x)$ and $y = g(x)$ intersect at $x = a$, $x = c$ and $x = b$ such that $a < c < b$. If $f(x) > g(x)$ in $[a, c]$ and $f(x) < g(x)$ in $[c, b]$, Then the area of the regions bounded by curve.



$$= \text{Area of the region PAQCP} + \text{Area of the region QDRBQ} = \int_a^c (f(x) - g(x)) dx + \int_c^b (g(x) - f(x)) dx$$